# Shaurya Agarwal

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## Education

**University of Washington** — Seattle, WA
*Master of Science in Data Science* — *Sep 2026 – Jun 2028 (Expected)*

**BITS Pilani, K. K. Birla Goa Campus** — Goa, India
*B.E. Computer Science; Minor in Data Science* — *Aug 2020 – Jun 2024*

## Work Experience

**Quantitative Analytics Specialist, Data Science** — Jul 2024 – Aug 2026
*Wells Fargo (promoted from Program Associate, Jul 2025)* — *Bengaluru, India*

- Developed semantic search and retrieval evaluation for an Elasticsearch-powered RAG assistant serving 9.1K daily bankers, replacing keyword-based procedure lookup and reducing procedure-related call volume by 12% after national rollout
- Isolated a ∼36% first-year drop in in-branch and ∼25% in digital account openings across 511 branch closures (Q1 2022–Q2 2025) against comparable still-open branches, estimating the ATT with channel-specific T-learner uplift models on block-group and CBSA features
- Built the national branch network simulator scoring ∼4,000 branches quarterly, forecasting account-opening loss for any candidate closure with gradient-boosted regression, backtested against realized outcomes and adopted by the Branch Systems & Transformation team
- Led the team that quantified a 145% brokerage balance lift for enrollees in the LifeSync in-app goal-planning feature among 17M deposit customers, using propensity score matching against comparable non-enrollees
- Doubled field-visit hit rate across ∼4,200 branches by rebuilding the Branch Dissatisfaction Index, forecasting six-month employee churn with a Cox proportional hazards model on sentiment and topic-model features from survey free-text

**Data Science Intern, Legal & Compliance** — Feb 2024 – Jun 2024
*Morgan Stanley* — *Mumbai, India*

- Stress-tested a developmental XGBoost insider-trading model built on client trading-behavior signals, documenting its detection coverage and failure modes across constructed test scenarios
- Cut false-positive alerts by ∼20% in manual compliance review, tuning detection thresholds the surveillance team adopted into live monitoring across EMEA and APAC typologies
- Automated weekly regression checks by replaying historical trade data through the surveillance platform, extending coverage with synthetic scenarios

## Projects

**Research Navigator, LLM Literature Synthesis** — github.com/Shaurya6355/research-navigator

- Designed an 8-stage literature research pipeline on the Anthropic SDK with schema-constrained JSON output, verbatim quote verification that drops unsupported claims, and prompt-injection delimiting, so every generated research direction traces to a source passage
- Benchmarked the pipeline under rubric-based LLM judging, grading 62 of 62 extracted claims genuine at 0.98 full support, and compared five retrieval and ranking strategies on nDCG@10 and precision@10 over 175 pooled pairs from 8 questions

## Teaching & Leadership

- Teaching Assistant for Principles of Programming Languages (cohort of 200) and Instructor at the campus Machine Learning Club (∼50 students) at BITS Pilani, designing and delivering ML and deep learning lectures and labs

## Skills

**Programming**: Python, C++, Java, SQL (Teradata), R
**Generative AI & Machine Learning**: retrieval-augmented generation (RAG), LLM application engineering, LLM evaluation, embeddings, semantic search, retrieval evaluation, natural language processing, causal inference (uplift modeling, propensity score matching), survival analysis (Cox proportional hazards), A/B testing and experimental design, scikit-learn, XGBoost, PyTorch, TensorFlow
**Data & Tools**: Anthropic SDK, Elasticsearch, vector search, pandas, NumPy, matplotlib, Jupyter, PySpark, PostgreSQL, GCP, BigQuery, FastAPI, Docker, MLflow, GitHub Actions, Git, Claude Code, GitHub Copilot